P3.4

(a) $\langle f|(\hat{A}+\hat{B})g\rangle = \langle f|\hat{A}g\rangle + \langle f|\hat{B}g\rangle = \langle \hat{A}f|g\rangle + \langle \hat{B}f|g\rangle =$
$= \langle (\hat{A}+\hat{B})f|g\rangle$

(b) $\langle f|\alpha\hat{Q}g\rangle = \alpha\langle f|\hat{Q}g\rangle = \alpha\langle \hat{Q}f|g\rangle$
$\langle \alpha\hat{Q}f|g\rangle = \alpha^*\langle \hat{Q}f|g\rangle$
$\alpha\hat{Q}$ is Hermitian for REAL $\underline{\alpha=\alpha^*}$

(c) $\langle f|\hat{A}\hat{B}g\rangle = \langle \hat{A}f|\hat{B}g\rangle = \langle \hat{B}\hat{A}f|g\rangle$

$\Rightarrow$ $\hat{A}\hat{B}$ is Hermitian if $\hat{A}\hat{B} = \hat{B}\hat{A} \Rightarrow [\hat{A},\hat{B}]=0$.

(d) $\langle f|\hat{x}g\rangle = \int f^* x g\, dx = \int (xf)^* g\, dx = \langle \hat{x}f|g\rangle$

$\hat{x}$ = Hermitian

$$\langle f|\hat{H}g\rangle = \int f^*\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)\right) g\, dx =$$

$$= \int f^*\left(-\frac{\hbar^2}{2m}\right)\frac{d^2 g}{dx^2}\, dx + \underbrace{\int f^* V(x) g\, dx}_{\int (V(x)f)^* g\, dx \text{ for } \underline{V(x)=\text{real}}}$$

$$\left(-\frac{\hbar^2}{2m}\right)\left\{\int f^*\frac{d^2 g}{dx^2}dx = f^*\frac{dg}{dx}\Big|_{-\infty}^{\infty} - \int\frac{df^*}{dx}\frac{dg}{dx}dx = \underbrace{\left(f^*\frac{dg}{dx} - \frac{df^*}{dx}g\right)\Big|_{-\infty}^{+\infty}}_{0 \text{ by boundary conditions}} + \int\frac{d^2 f^*}{dx^2}g\, dx\right\}$$

$f(x), g(x) \to 0$ $|x|\to\infty$

$$\Rightarrow \quad -\frac{\hbar^2}{2m}\int f^*\frac{d^2 g}{dx^2}dx = -\frac{\hbar^2}{2m}\int\frac{d^2 f^*}{dx^2}g\, dx$$

$\Rightarrow \langle f|\hat{H}g\rangle = \langle \hat{H}f|g\rangle \Rightarrow \hat{H}$ = Hermitian

$\boxed{P3.7}$ (a) $\left.\begin{array}{l}\hat{Q} f = q f \\ \hat{Q} g = q g\end{array}\right\}$ then $\hat{Q}(af + bg) = a\hat{Q}f + b\hat{Q}g =$

$= aqf + bqg = q(af + bg)$

for any coefficients $a, b$.

(b) Let $\hat{Q} = \frac{d^2}{dx^2}$, $f(x) = e^x$, $g(x) = e^{-x}$

$$\hat{Q} f = \frac{d^2 e^x}{dx^2} = e^x \Rightarrow q = 1$$

$$\hat{Q} g = \frac{d^2 e^{-x}}{dx^2} = (-1)^2 e^{-x} \Rightarrow q = 1$$

the same eigenvalue!

To construct orthogonal combinations on symmetric interval $(-1, 1)$

we just need to construct <u>even</u> and <u>odd</u> ones:

for example, $h = \frac{1}{2}(f + g) \Rightarrow h(-x) = h(x)$

$$t = \frac{1}{2}(f - g) \Rightarrow t(-x) = -t(x)$$

$h(x) = \cosh(x)$, $t(x) = \sinh(x)$.

and $\int_{-1}^{1} \cosh(x) \sinh(x)\, dx = 0$

$\boxed{P3.9}$ (a) Infinite square well

(b) Delta-function barrier $V(x) = +\alpha\delta(x)$.

(c) Delta-function potential well $V(x) = -\alpha\delta(x)$.

$\boxed{P3.11}$ Ground state w.f. of harmonic oscillator

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2} \qquad \text{eq. } (2.59)$$

$$\Rightarrow \Psi_0(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} e^{-i\left(\frac{\hbar\omega}{2}t\right)/\hbar}$$

Momentum space w.f.

$$\phi(p,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar}\, \psi(x,t)\, dx$$

$$\Rightarrow \phi_0(p,t) = \frac{1}{\sqrt{2\pi\hbar}} e^{-i\omega t/2} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \int_{-\infty}^{\infty} e^{-ipx/\hbar} e^{-\frac{m\omega}{2\hbar}x^2} dx$$

Complete square

$$\frac{m\omega}{2\hbar}x^2 + \frac{ipx}{\hbar} = \frac{m\omega}{2\hbar}\left(x^2 + i\,\frac{2}{m\omega}\,px\right) =$$

$$= \frac{m\omega}{2\hbar}\left[\left(x + \frac{ip}{m\omega}\right)^2 - \left(\frac{ip}{m\omega}\right)^2\right] = \frac{m\omega}{2\hbar}\left(x + i\frac{p}{m\omega}\right)^2 + \frac{p^2}{2\hbar m\omega}$$

$$\Rightarrow \phi_0(p,t) = \frac{e^{-i\omega t/2}}{\sqrt{2\pi\hbar}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \underbrace{\int_{-\infty}^{\infty} dx\, e^{-\frac{m\omega}{2\hbar}\left(x + i\frac{p}{m\omega}\right)^2}}_{\sqrt{\frac{\pi\cdot 2\hbar}{m\omega}}} e^{-\frac{p^2}{2\hbar m\omega}}$$

$$= \frac{1}{(\pi\hbar m\omega)^{1/4}} e^{-\frac{p^2}{2\hbar m\omega}} e^{-i\omega t/2}.$$

For a given $E_0 = \frac{1}{2}\hbar\omega$ classical range of momentum follows from positivity of $p^2$:

$$\frac{1}{2}\hbar\omega = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

$$\Rightarrow \quad p^2 < 2m \cdot \frac{1}{2}\hbar\omega = \hbar m\omega = p_0^2$$

Thus classically: $-p_0 < p < p_0 = \sqrt{\hbar m\omega}$

OR $\quad |p| < \sqrt{\hbar m\omega}$

Probability to measure $|p| > \sqrt{\hbar m\omega}$ is

$$P = \int_{p_0}^{\infty} |\phi(p,t)|^2 \, dp + \int_{-\infty}^{-p_0} |\phi(p,t)|^2 dp =$$

$$= 2 \int_{\sqrt{\hbar m\omega}}^{\infty} \frac{1}{(\pi \hbar m\omega)^{1/2}} e^{-\frac{p^2}{\hbar m\omega}} dp =$$

$$\left(p = \sqrt{\hbar m\omega}\,\eta\right) \quad \left(dp = \sqrt{\hbar m\omega}\, d\eta\right) \qquad = \frac{2}{\sqrt{\pi}} \underbrace{\int_1^{\infty} e^{-\eta^2} d\eta}_{\frac{\sqrt{\pi}}{2}\,\mathrm{Erfc}(1)} = \mathrm{Erfc}(1) = 0.1573 \approx 0.16$$

(the same as probability to find $x$ outside the classical range! P2.15)

Complimentary Error function $\mathrm{Erfc}(x) = 1 - \mathrm{Erf}(x)$

Error function

where $\mathrm{Erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$

$\boxed{\text{P3.13}}$

(a) $[AB, C] = ABC - CAB = ABC + ACB - ACB - CAB =$

$= A(BC - CB) + (AC - CA)B =$

$= A[B,C] + [A,C]B$

(b) Show $[x^n, p] = i\hbar n x^{n-1}$ $\qquad p = -i\hbar \frac{d}{dx}$

Take test function $f(x)$:

$$[x^n, p] f(x) = (-i\hbar)\left\{\left(x^n \frac{d}{dx} - \frac{d}{dx} x^n\right) f(x)\right\} =$$

$$= (-i\hbar)\left\{x^n \frac{df}{dx} - \frac{d}{dx}(x^n f)\right\} = (-i\hbar)\left\{x^n \frac{df}{dx} - n x^{n-1} f - x^n \frac{df}{dx}\right\}$$

$$= (-i\hbar)\{-n x^{n-1} f\} = i\hbar n x^{n-1} f(x)$$

$\Rightarrow [x^n, p] = i\hbar n x^{n-1}$ always.

(c) $[f(x), p] = i\hbar \frac{df}{dx}$

Take test function $g(x)$:

$$[f(x), -i\hbar \tfrac{d}{dx}]\, g(x) = (-i\hbar)\left(\left(f(x)\frac{d}{dx} - \frac{d}{dx}f(x)\right)g(x)\right)$$

$$= -i\hbar\left(f(x)\frac{dg}{dx} - \frac{d}{dx}(fg)\right) = -i\hbar\left(f\frac{dg}{dx} - f\frac{dg}{dx} - \frac{df}{dx}g\right)$$

$$= i\hbar\frac{df}{dx}\, g$$

$\Rightarrow$ always $[f(x), p] = i\hbar \frac{df}{dx}$.

$\boxed{P3.30}$ $\psi(x,0) = \frac{A}{x^2+a^2}$

(a) Normalize $1 = A^2 \int_{-\infty}^{\infty} \frac{1}{(x^2+a^2)^2}\,dx = (x = ay) = 2A^2 \frac{a}{a^4}\int_0^{\infty} \frac{dy}{(1+y^2)^2}$

$$= \frac{2A^2}{a^3}\frac{\pi}{4} \Rightarrow A = \sqrt{\frac{2a^3}{\pi}}$$

(b) $\langle x\rangle = A^2 \int_{-\infty}^{\infty} \frac{x}{(x^2+a^2)^2}\,dx = 0$ by oddness

$$\langle x^2\rangle = A^2 \int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)^2}\,dx = A^2 \int \frac{x^2+a^2-a^2}{(x^2+a^2)^2}\,dx =$$

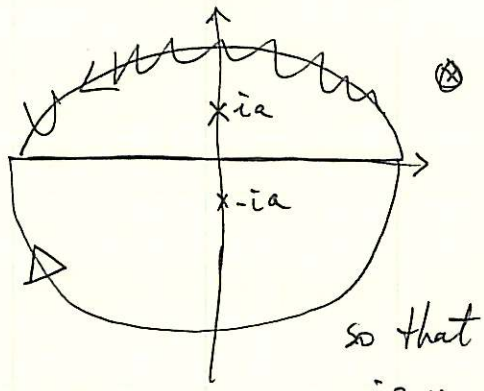
$$= A^2 \underbrace{\int_{-\infty}^{\infty} \frac{dx}{x^2+a^2}}_{2\cdot\frac{\pi}{2a}} - A^2a^2 \underbrace{\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2}}_{a^2\cdot 1} = \frac{\pi A^2}{a} - a^2 = \frac{\pi}{a}\frac{2a^3}{\pi} - a^2 = a^2.$$

$$\Rightarrow \sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \sqrt{a^2 - 0} = a.$$

(c) $$\phi(p,0) = \frac{A}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{x^2+a^2}\, dx$$

By Cauchi's theorem: $$I = \int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{x^2+a^2}\, dx = \int_{-\infty}^{\infty} \frac{e^{-ipx/\hbar}}{(x+ia)(x-ia)}\, dx$$

let $a>0$




$$= -2\pi i \, \frac{e^{-i\frac{p}{\hbar}(-ia)}}{-2ia} = \frac{\pi}{a} e^{-pa/\hbar}$$

so that

$e^{-i\frac{p}{\hbar}x} \to 0$ on semi-arc of infinite radius $(\operatorname{Im} X < 0)$

$$\Rightarrow \phi(p,0) = \frac{A}{\sqrt{2\pi\hbar}} \frac{\pi}{a} e^{-pa/\hbar} = \sqrt{\frac{2a^3}{\pi \cdot 2\pi\hbar} \frac{\pi^2}{a^2}}\, e^{-pa/\hbar} =$$

$$= \sqrt{\frac{a}{\hbar}}\, e^{-pa/\hbar}.$$ Clearly $p$ is positive here $p \to |p|$ in general.

(d) $$\langle p \rangle = \frac{a}{\hbar} \int_{-\infty}^{\infty} p\, e^{-2|p|a/\hbar}\, dp = 0$$

$$\langle p^2 \rangle = \frac{a}{\hbar} \int_{-\infty}^{\infty} p^2 e^{-2|p|a/\hbar}\, dp = \frac{2a}{\hbar} \int_0^{\infty} p^2 e^{-2pa/\hbar}\, dp$$

Call $\alpha = \frac{2a}{\hbar}$

$$\langle p^2 \rangle = \frac{2a}{\hbar} \int_0^{\infty} \frac{d^2}{d\alpha^2} e^{-\alpha p}\, dp = \frac{2a}{\hbar} \frac{d^2}{d\alpha^2} \frac{1}{\alpha} = \frac{2a}{\hbar} \frac{2}{\alpha^3} =$$

$$= \frac{2\hbar^2}{(2a)^2} = \frac{\hbar^2}{2a^2}$$

$$\Rightarrow \sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \frac{\hbar}{\sqrt{2}\, a}$$

(e) $$\sigma_x \sigma_p = a \frac{\hbar}{\sqrt{2}\, a} = \frac{\hbar}{\sqrt{2}} = \sqrt{2}\, \frac{\hbar}{2} > \frac{\hbar}{2}.$$